

Natural-language description

Urban air-quality monitoring with **3 drones** and **3 pollution sensing zones** over **3 time steps**.

Each robot can serve at most one task per time step. Some tasks require multiple robots simultaneously.

Each task must be completed within its valid time window.

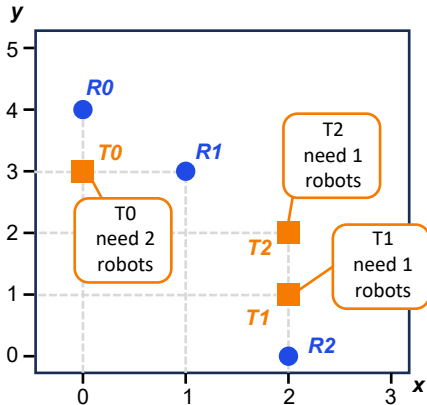
Objective: **minimize total Euclidean travel distance**.

(a)

Structured input/output

```
"input": {  
  "RobotPositions": [  
    [0.0, 4.0],  
    [1.0, 3.0],  
    [2.0, 0.0],  
  ],  
  "TaskPositions": [  
    [0.0, 3.0],  
    [2.0, 1.0],  
    [2.0, 2.0],  
  ],  
  "EarliestStep": [0, 0, 1],  
  "LatestStep": [1, 2, 2],  
  "NumRobots": 3,  
  "NumTasks": 3,  
  "TimeSteps": 3,  
  "TaskRequirements": [2, 1, 1]},  
"output": [4.0]
```

(b)



(c)